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IEEE Access - Decision on Manuscript ID Access-2026-35630

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

3 August 2026 at 12:41

Reply-To: akleylek@gmail.com

To: chettanat.pa@gmail.com, surawutw@g.swu.ac.th, sueb@pnru.ac.th, prathan.bu@kmitl.ac.th

Cc: akleylek@gmail.com

03-Aug-2026

Dear Dr. Wicharn:

I am writing to you regarding manuscript # Access-2026-35630 entitled "Design and Simulation of a Broadband Tunable Transmittance Optical Filter Using Sb₂Se₃ Phase Change Material" which you submitted to IEEE Access.

Your article was peer reviewed with interest but has not been recommended for publication in its current form. We strongly encourage you to address the reviewers' concerns, which can be found at the bottom of this letter, and resubmit your article to IEEE Access once you have updated it accordingly.

Please note that IEEE Access has a binary peer review process. Therefore, to uphold quality to IEEE standards, an article is rejected even if it requires minor edits.

When updating your manuscript, you should elaborate on your points and clarify with references, examples, data, etc. If you disagree with any technical points the reviewers have made, please include your counterarguments in your response to the reviewers (more information detailed below) and work this into the updated manuscript.

Also, note that if a reviewer suggested references, you should only add those that are relevant to your work if you feel they strengthen your article. Recommending references to specific publications is not appropriate for reviewers and you should report excessive cases to ieeeaccessEIC@ieee.org. Authors are not obligated to cite articles that are recommended by the reviewers, and the final decision on the article will not be influenced by whether or not authors cite these suggested references.

IEEE Access allows one opportunity to resubmit. If the updated manuscript is determined not to have addressed all of the previous reviewers' concerns, or if the Associate Editor still has substantial technical concerns, the article will be rejected and no further resubmissions will be allowed.

When you are ready to resubmit your updated article, you can do so in the [IEEE Author Portal](#). When you log into the IEEE Author Portal you will see the title of the rejected article and the option to "Start Resubmission".

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Upon resubmission you will be asked to upload the following 3 files:

- 1) A document containing your response to reviewers from the previous peer review. The "response to reviewers" document (template attached) should have the following regarding each comment: a) Reviewer's concern, b) your response to the concern, c) your action to remedy the concern. **The document should be uploaded with your manuscript files under "Author's Response Files."**
- 2) Your updated manuscript with all your individual changes highlighted, including grammatical changes (e.g. preferably with the yellow highlight tool within the pdf file). This file should be uploaded with your manuscript files as *"Highlighted PDF."*
- 3) A clean copy of the final manuscript (without highlighted changes) submitted as a Word or LaTeX file, and as a PDF, both submitted as the *"Main Manuscript."*

****IMPORTANT:** Please see the attached Resubmission Checklist that details all the items listed above. Please utilize this checklist to ensure you have made the necessary edits to your manuscript, and to ensure you have all the necessary files prepared prior to resubmission.

***** AUTHOR LIST CHANGES:** If your revised manuscript has an updated author list, you will need to submit a formal request to the Editor by completing the attachment labelled 'Request for Byline Change,' and uploading it as *'Request for byline change form.'* This should include a DETAILED justification explaining each author's contribution(s) to the

work. You will also need to provide the justification for the author change during the submission process. Change in the author list is considered rare and exceptional, and the decision to allow such changes rests with the Editor. Once the list and order of authors has been established, the list and order of authors should not be altered without permission of all living authors of that article.

We sincerely hope you will update your manuscript and resubmit soon. Please contact me if you have any questions.

Thank you for your interest in IEEE Access.

Sincerely,

Dr. Sedat Akleylek
Associate Editor, IEEE Access
akleylek@gmail.com

Reviewers' Comments to Author:

Reviewer: 1

Comments:

1. Clarify the novel and innovative aspects of the work by comparing the proposed design with state-of-the-art Sb_2Se_3 -based optical filters.
2. Support simulation results with experimental evidence or published experimental data.
3. Provide a stronger justification for the simulation parameters and assumptions employed.
4. Include a comparison table featuring currently available state-of-the-art optical filters.
5. Validate the endurance model using experimental data or stronger evidence from the scientific literature.
6. Discuss the feasibility of practical fabrication and implementation challenges.
7. Assess the impact of fabrication tolerances on device performance.
8. Strengthen the discussion section by providing a deeper analysis of the simulation results.
9. Expand the conclusion section to include limitations and future research directions.

Additional Questions:

Please confirm that you have reviewed all relevant files, including supplementary files and any author response files, which can be found in the "View Author's Response" link above (author responses will only appear for resubmissions):
Yes, all files have been reviewed

- 1) Does the paper contribute to the body of knowledge?: Yes
- 2) Is the paper technically sound?: Yes
- 3) Is the subject matter presented in a comprehensive manner?: Yes
- 4) Are the references provided applicable and sufficient?: Yes, the references are applicable, but they are not sufficient
- 5) Are there references that are not appropriate for the topic being discussed?: No
- 5a) If yes, then please indicate which references should be removed.:

Reviewer: 2

Comments:

The paper presents a TMM-based simulation study comparing single-, two-, three-layer, and long-pass edge-filter thin-film designs using Sb_2Se_3 as a tunable optical filter at 1064 nm. A 5-unit-cell edge filter is selected as the best compromise (93.4% ON, 0.2% OFF transmittance), with supplementary analyses of endurance over 1.5×10^6 cycles, angle/polarization dependence, and a merit-function optimization. The work is a purely computational/simulation study with no experimental fabrication or validation.

1. Clarify the novelty by explicitly distinguishing the proposed symbol-level constant-envelope waveform design from existing constant-envelope OTFS and CPM-based ISAC approaches in the Introduction and Related Work.
2. The entire study is TMM simulation; there is no fabricated sample, measured spectrum, or comparison to any prior experimental Sb_2Se_3 filter to anchor the modeled numbers (transmittance, ΔT , endurance). Given IEEE Access's broad but rigor-focused scope, at minimum a sensitivity/tolerance analysis (e.g., ± 5 – 10% layer thickness error) would strengthen practical credibility.
3. The paper repeatedly claims benefits like "less Sb_2Se_3 used... minimizes energy required for phase transition" (Sec. IV.A) and discusses electrical vs. optical heating for the multilayer configurations (Sec. 4.3), but no switching energy, power, or thermal simulation is provided to substantiate these practical tunability claims.

4. $\lambda_0 = 740$ nm is chosen "as an example" to align the AR/HR edge with 1064 nm, and only afterward is an optimizer applied (Sec. 4.3). Explaining why 740 nm specifically (vs. a systematically swept λ_0) would improve rigor; the current framing suggests trial-and-error justified post hoc.
5. Fig. 7 compares only N=4 and N=5 edge filters; the final chosen design is N=5, but no rationale is given for omitting N=3 or N=6 from this specific comparison, especially since N=6 has the highest ΔT reported in Table I.
6. Improve reproducibility by providing additional simulation details (e.g., parameter settings, initialization, convergence criteria) and indicate whether the simulation code will be publicly available.
7. Table I column header "SbSe used (nm)" should read "Sb₂Se₃ used (nm)"; also clarify FWHM computation method (measured on the ΔT spectrum vs. individual Ta/Tc curves) since this metric is quoted precisely (197.0 nm, 207.4 nm) without defined methodology. Minor presentation improvements are recommended, including proofreading for notation consistency, ensuring all variables are defined upon first use, and improving the readability of several equations and figures.

Additional Questions:

Please confirm that you have reviewed all relevant files, including supplementary files and any author response files, which can be found in the "View Author's Response" link above (author responses will only appear for resubmissions):
Yes, all files have been reviewed

- 1) Does the paper contribute to the body of knowledge?: Yes
- 2) Is the paper technically sound?: Largely yes for the core TMM optics
- 3) Is the subject matter presented in a comprehensive manner?: Reasonably comprehensive for a simulation study, it covers material properties, multiple design architectures, angle/polarization robustness, cycling endurance, and tunability/optimization. Missing: switching energy/actuation feasibility, fabrication tolerance sensitivity, and any real-world benchmarking.
- 4) Are the references provided applicable and sufficient?: Yes
- 5) Are there references that are not appropriate for the topic being discussed?: No
- 5a) If yes, then please indicate which references should be removed.:

Reviewer: 3

Comments:

This manuscript presents a promising Sb₂Se₃-based tunable filter design with high simulated performance. However, the work is purely numerical, lacks experimental validation or fabrication discussion, and omits a comparative benchmark with existing devices. Therefore, I recommend a major revision. My specific comments are as follows,

1. This work is simulation-only. If possible, include experimental validation; otherwise, discuss fabrication feasibility.
2. Add a benchmark comparison with existing tunable filters at the end of Section IV to better highlight the advantages of this design.
3. Figure 7 shows that beyond 30° incidence, contrast degrades significantly for both polarizations. Specify 0°–30° as the effective working range and briefly explain the physical cause.
4. Add a schematic diagram of the interference mechanism in Section II to improve readability.
5. Unify the legend styles across all subfigures in Figure 8.
6. Equations (14) and (15) are repetitive; combine or simplify them.

Additional Questions:

Please confirm that you have reviewed all relevant files, including supplementary files and any author response files, which can be found in the "View Author's Response" link above (author responses will only appear for resubmissions):
Yes, all files have been reviewed

- 1) Does the paper contribute to the body of knowledge?: Yes.
- 2) Is the paper technically sound?: Yes.
- 3) Is the subject matter presented in a comprehensive manner?: Yes.
- 4) Are the references provided applicable and sufficient?: Yes.
- 5) Are there references that are not appropriate for the topic being discussed?: No
- 5a) If yes, then please indicate which references should be removed.:

Reviewer: 4

Comments:

The manuscript presents an interesting optical modeling for multilayer Sb₂Se₃ thin-film filter design using the transfer matrix method. However, many of the claimed device-level conclusions are inferred rather than demonstrated.

1. The paper claims a broadband tunable optical filter, yet the authors manually assign different phase configurations inside the simulation, which makes the idea of continuous tuning and wavelength tuning difficult to experimentally realize. The author demonstrates many possible optical states but not necessarily a realizable tunable optical filter.
2. This paper mostly changes transmission amplitude instead of substantially shifting the filter spectrum. The optimization only slightly modifies the spectrum ($\sim < 50$ nm). That is optimization, not significant wavelength tunability.
3. Tables I and II optimize layer thicknesses on the $\sim$ nm scale using classical thin-film interference. The paper assumes bulk refractive index and extinction coefficient are independent of thickness. The use of bulk values should be justified for those that could have arisen from finite-thickness effects, interface quality, and crystallization non-uniformity.
4. Fig. 8 demonstrates propagation of assumed material degradation. The endurance simulation follows simple exponential and linear drift equations, whose coefficients are taken from literature, rather than derived from phase-transition simulations.
5. For 93.6% transmission, the insertion loss is $\sim IL = -10 \log_{10}(0.936) \sim 0.3$ dB. However, Fig. 8 reports insertion loss evolution relative to the initial state, not absolute insertion loss. The authors should also report: a) initial IL, b) final IL, and c) absolute IL. Also, for non-monotonic behavior observed in 4 layers $\Rightarrow$ 5 $\Rightarrow$ 6, no systematic explanation is given. Justification given through field confinement and resonance is not explained properly for this non-monotonic behavior.
6. Motivation of the presented work is missing in the introduction and abstract. The authors state Sb₂Se₃ has not been explored much, but the paper cites recent Sb₂Se₃ optical filter studies and does not quantitatively compare against them. How does this paper stand out from recent work? 1) <https://iopscience.iop.org/article/10.1088/2040-8986/ad80a6/meta>; 2) <https://opg.optica.org/ome/fulltext.cfm?uri=ome-12-11-4268>; 3) <https://opg.optica.org/oe/fulltext.cfm?uri=oe-34-5-7378>

Why another filter? What gap exists? What limitation exists? What application cannot currently be achieved? A comparison table against the works needed to strengthen the novelty claim and strengthen the introduction.

Additional Questions:

Please confirm that you have reviewed all relevant files, including supplementary files and any author response files, which can be found in the "View Author's Response" link above (author responses will only appear for resubmissions):
Yes, all files have been reviewed

- 1) Does the paper contribute to the body of knowledge?: Paper contribution is incremental. The main contribution of the paper is a simulation based design methodology rather than a new optical filter.
- 2) Is the paper technically sound?: Several assumptions reduce technical rigor like a) optical constants are assumed to remain bulk values regardless of layer thickness, b) endurance is modeled through assumed empirical drift equations rather than physical degradation mechanism, etc. These assumptions should be acknowledged more explicitly, and some conclusions are stronger than the supporting evidence.
- 3) Is the subject matter presented in a comprehensive manner?: Partially comprehensive.
This paper does not discuss the novelty, benchmarking : a) why this study is needed relative to existing literature, b) quantitative comparison with previous Sb₂Se₃ filter designs, c) sensitivity to thickness tolerances, d) modulation depth, e) absolute insertion loss, etc.
- 4) Are the references provided applicable and sufficient?: References cited are applicable but the discussion of prior work and differentiation from existing literature should be strengthened, like:
M. Cui et al., Tunable multi-channel optical filter based on the near-infrared ultralow-loss phase change material Sb₂Se₃, Optics Express, 34(5), 7378–7387 (2026).

DOI: 10.1364/OE.586098

5) Are there references that are not appropriate for the topic being discussed?: No

5a) If yes, then please indicate which references should be removed.:

Reviewer: 5

Comments:

This paper designs and simulates a switchable 1064 nm near-infrared optical filter using the phase-change material

Sb₂Se₃, achieving over 93% transmittance contrast and endurance of roughly one million switching cycles. This manuscript falls well within IEEE Access's scope. Here are some suggestions.

Q1:

In the abstract,

- 1) please consider adding one sentence on why a switchable filter at this wavelength is needed and what limits existing PCM options.
- 2) if the work is purely simulation-based, use language like "simulation results show" rather than definitive statements, to accurately reflect the study's nature.

Q2:

The Introduction spends considerable space defining PCM and its classifications (volatile vs. non-volatile, VO₂'s transition mechanism, etc.), which may be more detail than necessary for this journal's readership. More importantly, I could not find a clear, explicit statement of what specific problem or limitation in existing PCM-based tunable filters this work aims to solve—could the authors clarify what practical shortcomings of current materials/designs (e.g., excessive loss, low endurance, narrow bandwidth) motivated this study, and state this more directly before introducing their proposed approach?

Q3:

The drift coefficient (α_p) and loss growth coefficient (β_p) in Equations 18–19 are stated to be "consistent with experimental trends," but no specific numerical values or their literature source are given. Could the authors clarify exactly which experimental dataset these coefficients were fitted to, and report the actual values used in the simulation?

Q4:

The paper states the 5-unit-cell design offers "the best compromise" between performance, size, and complexity, but this appears to be a qualitative judgment. Could the authors provide a quantitative figure-of-merit (e.g., a weighted score combining ΔT , thickness, and material usage) to justify why N=5 was chosen over N=4 or N=6, rather than N=6 which showed the highest ΔT ?

Q6:

Since all data are simulation-based and rely on literature-sourced refractive index values (which themselves carry measurement uncertainty), could the authors comment on how sensitive the reported ΔT , FWHM, and endurance values are to small variations (e.g., $\pm 1\text{--}2\%$) in the input n/k datasets?

Q7:

On the incident angle/polarization data (Fig. 7), The paper notes TM polarization causes a "significant blueshift" at higher incident angles but does not quantify this shift numerically. Could the authors report the specific magnitude of the blueshift (in nm) at, say, 45° and 60° for both the 4- and 5-unit-cell designs?

Additional Questions:

Please confirm that you have reviewed all relevant files, including supplementary files and any author response files, which can be found in the "View Author's Response" link above (author responses will only appear for resubmissions):
Yes, all files have been reviewed

- 1) Does the paper contribute to the body of knowledge?: Yes
- 2) Is the paper technically sound?: Moderate
- 3) Is the subject matter presented in a comprehensive manner?: Moderate
- 4) Are the references provided applicable and sufficient?: Yes
- 5) Are there references that are not appropriate for the topic being discussed?: No
- 5a) If yes, then please indicate which references should be removed.:

If you have any questions, please contact article administrator: Miss Manpreet Kaur m.kaur@ieee.org

3 attachments



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